

Project

AIM:

This project showcases key features and object-oriented programming (OOP) principles through a simulated banking system, employing encapsulation with a private balance attribute, methods for depositing, withdrawing, and checking balance, validation checks to prevent negative deposits and demonstrating OOP principle including class definition, constructor methods, instance methods, data validation, and object instantiation.

CODE:

```
class ATMSystem:
```

```
    def __init__(self):
```

```
        # Format: card_number : [PIN, balance]
```

```
        self.accounts = {
```

```
            "8809972": ["54321", 8000.0],
```

```
            "9155696": ["4321", 9000.0],
```

```
            "1234567": ["1144", 7000.0],
```

```
        }
```

```
    def authenticate(self, card_number, pin):
```

```
        if card_number in self.accounts and self.accounts[card_number][0] == pin:
```

```
            return True
```

```
        return False
```

```
    def deposit(self, card_number, amount):
```

```
        if amount <= 0:
```

```
            print("Deposit amount must be positive.")
```

```
        else:
```

```
            self.accounts[card_number][1] += amount
```

```
        print(amount , "deposited successfully.")

        balance = self.accounts[card_number][1]

        print("Current Balance:",balance)

def withdraw(self, card_number, amount):

    balance = self.accounts[card_number][1]

    if amount <= 0:

        print("Withdrawal amount must be positive.")

    elif amount > balance:

        print("Insufficient balance.")

    else:

        self.accounts[card_number][1] -= amount

        print(amount,"withdrawn successfully.")

        balance = self.accounts[card_number][1]

        print("Current Balance:",balance)

def check_balance(self, card_number):

    balance = self.accounts[card_number][1]

    print("Current Balance:",balance)

def run(self):

    print("Welcome to the ATM and BANK Account System")

    card_number = input("Enter Card Number: ")

    pin = input("Enter PIN: ")

    if not self.authenticate(card_number, pin):

        print("Invalid card number or PIN.")

        return

    while True:
```

```
print("\nMenu:")

print("1. Check Balance")

print("2. Deposit Money")

print("3. Withdraw Money")

print("4. Exit")

choice = input("Enter choice (1-4): ")

if choice == "1":

    self.check_balance(card_number)

elif choice == "2":

    amount = float(input("Enter amount to deposit: "))

    self.deposit(card_number, amount)

elif choice == "3":

    amount = float(input("Enter amount to withdraw: "))

    self.withdraw(card_number, amount)

elif choice == "4":

    print("Thank you for using the ATM!")

    break

else:

    print("Invalid choice. Please try again.")

# Run the ATM System

if __name__ == "__main__":

    atm = ATMSystem()

    atm.run()
```

OUTPUT:

```

Welcome to the ATM and BANK Account System
Enter Card Number: 8809972
Enter PIN: 54321

```

Menu:

1. Check Balance
2. Deposit Money
3. Withdraw Money
4. Exit

```

Enter choice (1-4): 2
Enter amount to deposit: 1000
1000.0 deposited successfully.
Current Balance: 9000.0

```

Menu:

1. Check Balance
2. Deposit Money
3. Withdraw Money
4. Exit

```

Enter choice (1-4): 3
Enter amount to withdraw: 500
500.0 withdrawn successfully.
Current Balance: 8500.0

```

Menu:

1. Check Balance
2. Deposit Money
3. Withdraw Money
4. Exit

```

Enter choice (1-4): 1
Current Balance: 8500.0

```